6 Principal Component Analysis

MA2003B Application of Multivariate Methods in Data Science

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Principal Component Analysis (PCA) is a powerful technique used to reduce the dimensionality of a dataset while preserving as much information (measured by variance) as possible.

It transforms the original variables into a new set of uncorrelated variables called **principal components,** which are ordered by the amount of variance they explain.

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Understanding Total Variance

Let Σ be the covariance matrix of a given dataset.

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The **total variance** of the data is given by the trace of the covariance matrix:

$$\text{Total Variance} = \operatorname{Tr}\Sigma = \sum_{i=1}^{p} \Sigma_{ii},$$

where Σ_{ii} are the diagonal elements of Σ representing the sample variance of each variable.

Now, using the singular value decomposition (SVD) of Σ, we can find matrices Q, and Λ such that:

$$\Sigma = Q\Lambda Q^T,$$

where Λ is a diagonal matrix containing the singular values $\lambda_1, \ldots, \lambda_p$ of Σ.

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where Λ is a diagonal matrix containing the singular values $\lambda_1, \ldots, \lambda_p$ of Σ.

These singular values can then be used to express the total variance as:

$$\text{Total Variance} = \operatorname{Tr}\Sigma = \operatorname{Tr} Q\Lambda Q^T = \operatorname{Tr}\Lambda = \lambda_1 + \cdots + \lambda_p.$$

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The first principal component (associated with the largest singular value) captures the most variance, the second principal component (associated with the second largest singular value) captures the second most variance, and so on.

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They are related to the original variables through the equations:

$$X = Q\,(PC) + \bar{x}$$

and

$$PC = Q^T\,(X - \bar{x})$$

Here X is our original coordinate vector, PC is the coordinate vector corresponding to the principal component scores, and $\bar{x}$ is the vector of means of our dataset.

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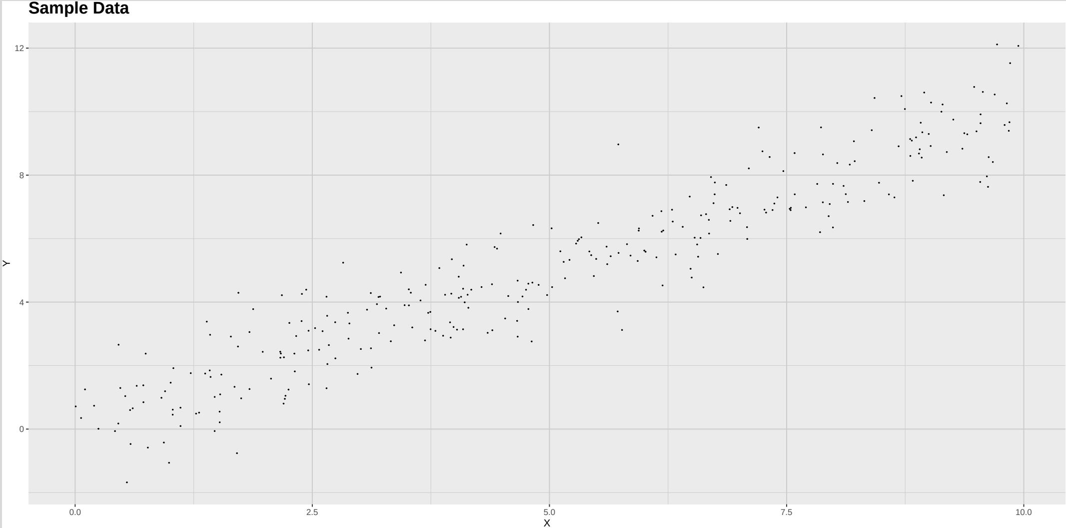
To illustrate these ideas, we will consider again the dataset shown in the following figure:


Sample Data
Y
12
8
4
0
0.0
2.5
5.0
7.5
10.0
X

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The second variable is generated as $Y = X + \varepsilon$, where ε is a normally distributed random error with mean 0 and standard deviation 1.

The covariance matrix of this dataset is given by:

$$\Sigma = \begin{bmatrix} 7.874751 & 7.7745959 \\ 7.7745959 & 8.6510962 \end{bmatrix}$$

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The total variance is:

$$\text{Total Variance} = \mathrm{Tr}\,\Sigma = \Sigma_{11} + \Sigma_{22} = 16.5258472$$

We now look at the singular value decomposition of Σ, to obtain:

$$Q = \begin{bmatrix} 0.689251 & -0.7245227 \\ 0.7245227 & 0.689251 \end{bmatrix}$$

and

$$\Lambda = \begin{bmatrix} 16.0472039 & 0 \\ 0 & 0.4786433 \end{bmatrix}$$

The principal component directions are given by the columns of Q, and the amount of variance explained by each component is given by the corresponding singular value in Λ.

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In particular, since the vector of means is given by:

$$\bar{x} = 4.9961185$$
$$\bar{y} = 5.019733$$

we have that the relation between the original variables and the principal component coordinates is given by:

$$X = 0.689251PC_1 + -0.7245227PC_2 + 4.9961185$$
$$Y = 0.7245227PC_1 + 0.689251PC_2 + 5.019733$$

and

$$PC_1 = 0.689251(X - 4.9961185) + 0.7245227(Y - 5.019733)$$
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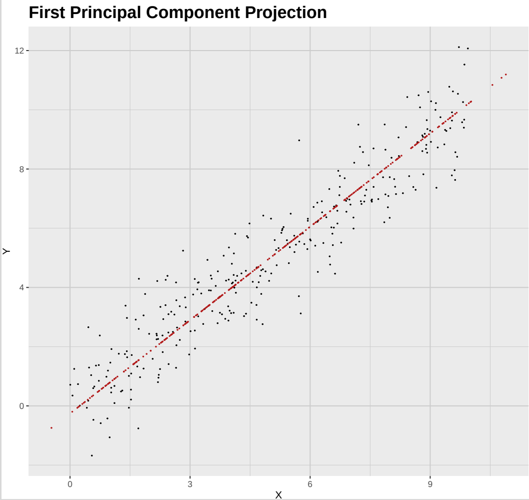
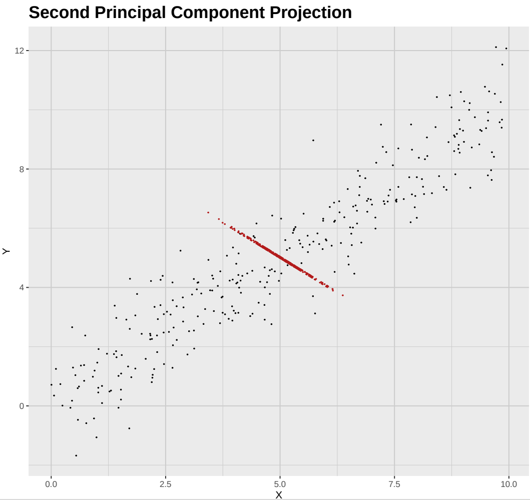
using these formulas, we can find the projection of our original data into the first and second principal components. The results are shown in the following figure:

Principal Component Projections





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To decide how many principal components to keep, we can look at the amount of variance explained by each component.

A common way to visualize this information is through the use of a **scree plot**.

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The scree plot associated to a given dataset displays the singular values on the y-axis and the component number on the x-axis.

It helps to identify the point where the explained variance starts to level off, often referred to as the "elbow".

Another useful plot is the **cumulative variance plot**, which shows the cumulative proportion of variance explained by the principal components.

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Notice that the proportion of variance explained by the k-th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^{p} \lambda_i}$$

This plot helps to determine how many components are needed to explain a desired amount of total variance, such as 70% or 90%.

To illustrate this, we will use the well-known **Iris** dataset, which contains measurements of four features:

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

For three species of iris flowers:

1. Iris setosa
2. Iris versicolor
3. Iris virginica

We will perform PCA on the four features and create a scree plot to visualize the variance explained by each principal component.

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The covariance matrices for each species are given by:

$$\Sigma_{setosa} = \begin{bmatrix} 0.12 & 0.1 & 0.02 & 0.01 \\ 0.1 & 0.14 & 0.01 & 0.01 \\ 0.02 & 0.01 & 0.03 & 0.01 \\ 0.01 & 0.01 & 0.01 & 0.01 \end{bmatrix}$$

$$\Sigma_{versicolor} = \begin{bmatrix} 0.27 & 0.09 & 0.18 & 0.06 \\ 0.09 & 0.1 & 0.08 & 0.04 \\ 0.18 & 0.08 & 0.22 & 0.07 \\ 0.06 & 0.04 & 0.07 & 0.04 \end{bmatrix}$$

$$\Sigma_{virginica} = \begin{bmatrix} 0.4 & 0.09 & 0.3 & 0.05 \\ 0.09 & 0.1 & 0.07 & 0.05 \\ 0.3 & 0.07 & 0.3 & 0.05 \\ 0.05 & 0.05 & 0.05 & 0.08 \end{bmatrix}$$

The singular values for each species are given by:

$$\Lambda_{setosa} = \begin{bmatrix} 0.24 & 0 & 0 & 0 \\ 0 & 0.04 & 0 & 0 \\ 0 & 0 & 0.03 & 0 \\ 0 & 0 & 0 & 0.01 \end{bmatrix}$$

$$\Lambda_{versicolor} = \begin{bmatrix} 0.49 & 0 & 0 & 0 \\ 0 & 0.07 & 0 & 0 \\ 0 & 0 & 0.05 & 0 \\ 0 & 0 & 0 & 0.01 \end{bmatrix}$$

$$\Lambda_{virginica} = \begin{bmatrix} 0.7 & 0 & 0 & 0 \\ 0 & 0.11 & 0 & 0 \\ 0 & 0 & 0.05 & 0 \\ 0 & 0 & 0 & 0.03 \end{bmatrix}$$

The associated principal component directions are given by the columns of the matrices

$$Q_{setosa} = \begin{bmatrix} -0.67 & 0.6 & 0.44 & -0.04 \\ -0.73 & -0.62 & -0.27 & -0.02 \\ -0.1 & 0.49 & -0.83 & -0.24 \\ -0.06 & 0.13 & -0.2 & 0.97 \end{bmatrix}$$

$$Q_{versicolor} = \begin{bmatrix} -0.69 & 0.67 & -0.27 & 0.1 \\ -0.31 & -0.57 & -0.73 & -0.23 \\ -0.62 & -0.34 & 0.63 & -0.32 \\ -0.21 & -0.34 & 0.06 & 0.92 \end{bmatrix}$$

$$Q_{virginica} = \begin{bmatrix} -0.74 & -0.17 & -0.53 & 0.37 \\ -0.2 & 0.75 & -0.33 & -0.54 \\ -0.63 & -0.17 & 0.65 & -0.39 \\ -0.12 & 0.62 & 0.43 & 0.65 \end{bmatrix}$$

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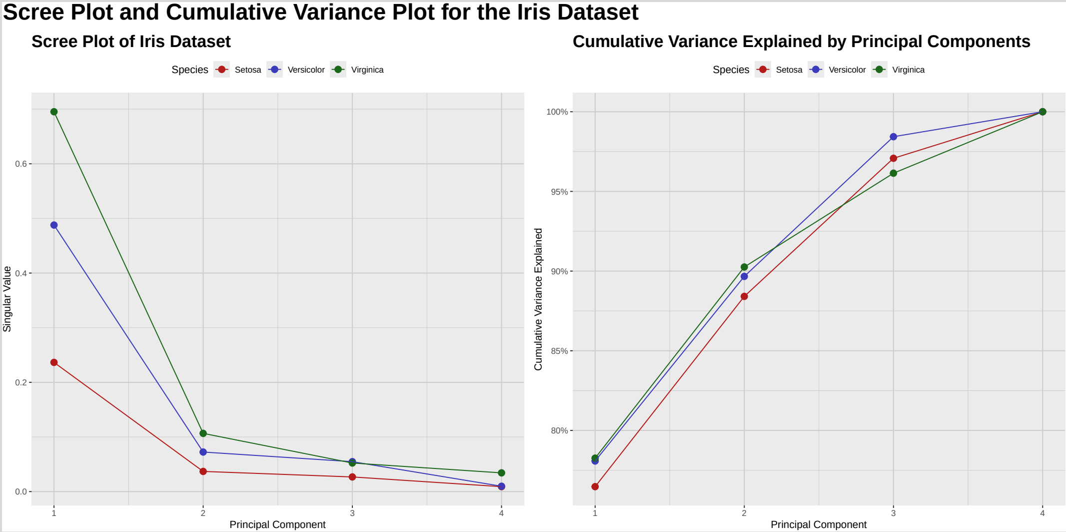
The scree plot and cumulative variance plot for the three species are shown in the following figure:


Scree Plot and Cumulative Variance Plot for the Iris Dataset
Scree Plot of Iris Dataset
Species Setosa Versicolor Virginica
Singular Value
0.0
0.2
0.4
0.6
1
2
3
4
Principal Component
Cumulative Variance Explained by Principal Components
Species Setosa Versicolor Virginica
Cumulative Variance Explained
80%
85%
90%
95%
100%
1
2
3
4
Principal Component

Loadings and Biplots

Recall that equation

$$PC = Q^T (X - \bar{x})$$

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Using this equation, we can quickly compute these scores for each observation in our dataset.

The coefficients appearing in this equation are known as the **loadings** of the principal components, and they indicate how much each original variable contributes to each principal component.

A **biplot** is a graphical representation that displays both the principal component scores of the observations and the loadings of the variables on the same plot.

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To illustrate this concept, we will create biplots for the first two principal components of the **iris** dataset for each species.

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This allows us to visualize how the original variables contribute to the principal components and how the observations are distributed in the space defined by these components.

To illustrate this concept, we will create biplots for the first two principal components of the **iris** dataset for each species.

We start by finding the loadings corresponding to each species, for which we simply take the first two columns of the matrices Q_{setosa}, $Q_{versicolor}$, and $Q_{virginica}$ shown above.

$$\text{Loadings}_{setosa} = \begin{bmatrix} -0.67 & 0.6 \\ -0.73 & -0.62 \\ -0.1 & 0.49 \\ -0.06 & 0.13 \end{bmatrix}$$

$$\text{Loadings}_{versicolor} = \begin{bmatrix} -0.69 & 0.67 \\ -0.31 & -0.57 \\ -0.62 & -0.34 \\ -0.21 & -0.34 \end{bmatrix}$$

$$\text{Loadings}_{virginica} = \begin{bmatrix} -0.74 & -0.17 \\ -0.2 & 0.75 \\ -0.63 & -0.17 \\ -0.12 & 0.62 \end{bmatrix}$$

We can then compute the principal component scores for each observation by simply multiplying the values of our original variables by these matrices.

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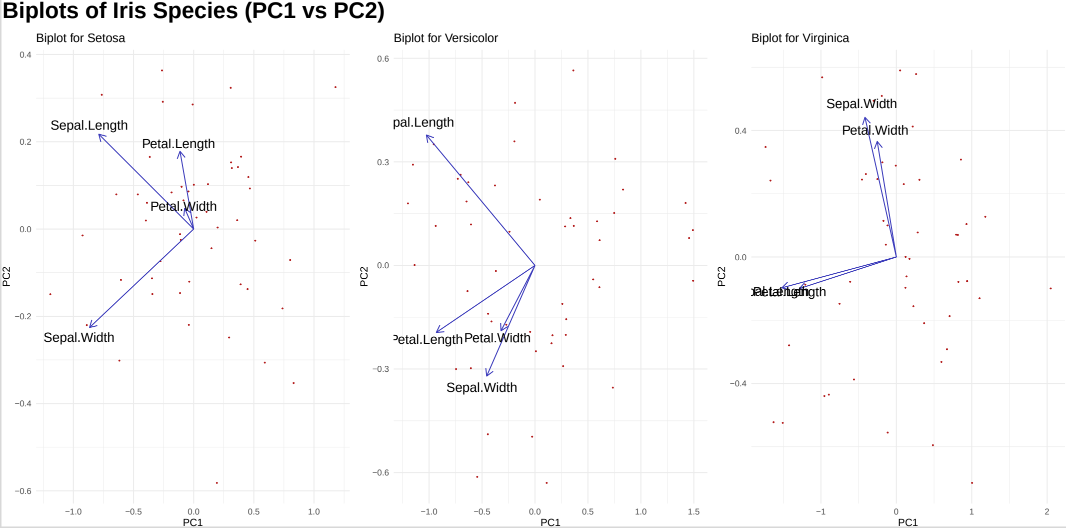
The resulting biplots for each species are shown in the following figure:


Biplots of Iris Species (PC1 vs PC2)
Biplot for Setosa
Sepal.Length
Petal.Length
Petal.Width
Sepal.Width
PC1
PC2
Biplot for Versicolor
pal.Length
Petal.Length
Petal.Width
Sepal.Width
PC1
PC2
Biplot for Virginica
Sepal.Width
Petal.Width
PC1
PC2

Summary of the PCA Process

In general, when performing PCA, it is recommended to follow these steps:

1. **Standardize the data (if needed).** It is not always easy to decide whether to standardize the data or not. Here are some guidelines:
    - If the variables are measured on different scales (e.g., height in cm and weight in kg), standardization is usually recommended.
    - If the variables are on the same scale and have similar variances, standardization may not be necessary.
    - If the variables have very different variances, standardization can help to give equal weight to all variables.

2. **Compute the covariance matrix.**
   - The matrix captures how variables vary together.
   - Its size is $p \times p$, where p is the number of variables.

3. **Find the singular value decomposition (SVD) of the covariance matrix.**
    - Eigenvalues measure the variance explained by each principal component.
    - Eigenvectors give the directions (linear combinations of variables) of the components.

4. **Rank the components.**
   - Order the singular values from largest to smallest.
   - The singular vector corresponding to the largest singular value is the first principal component.

5. **Decide how many components to keep.**
    - *Kaiser criterion*: keep the singular values > 1 (for correlation matrix).
    - *Scree plot*: look for the "elbow" where explained variance levels off.
    - *Cumulative variance*: keep enough components to explain about $70\% - 90\%$ of the total variance.

6. **Find the principal component scores and loadings.**
    - Project the original data onto the selected principal components.
    - Use the formula: $PC = Q^T(X - \bar{x})$.

7. **Interpret the components.**
    - Examine the *loadings* to see which variables contribute the most.
    - Larger absolute values indicate stronger influence.
    - Use biplots to visualize scores and loadings together.
    - Apply dimension reduction before regression, clustering, or other analyses.
    - Apply noise reduction by discarding low-variance components.